



# Enhancement of a heat sink performance using a partial shield and/or a guide plate

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## ABSTRACT

The potential of using a full shield or partial shield with a guide plate to control the bypass cooling air flowing above a heat sink is numerically modeled using two different scenarios of inlet conditions – fixed mass flow rate and specified fan curve. Excellent agreement is observed when the simulation results are compared with available experimental data in the literature. Under the first inlet condition, the Nusselt numbers associated with the full shield installation are significantly higher than those without a shield, but with a significant pressure drop penalty. Friction factor increases with partial shield height till it reaches a maximum value with the full-shield configuration. Using an inclined guide plate with a partial shield alters the flow direction, forcing the air to impinge on the heat sink. This adds an extra cooling effect but without causing a considerable increase in the friction factor. In the case of fixed inlet mass flow rate, most of the partial shield and guide plate cases show the lowest pressure drop when the ratio of the partial shield height to the total bypass height  $H_s/H_{BP}$  is between 0.35 and 0.5. The case with an inclination angle of  $\theta_p = 15^\circ$  has the highest thermal performance and the highest pressure drop. Correlations for the average Nusselt number and friction factor are developed to include the effect of Reynolds number, shield height, bypass height and guide plate inclination angle. For the second inlet condition, the test cases are repeated using a selected fan characteristic curve as an alternate to the fixed mass flow rate condition. The specified fan curve case shows that a short partial shield with a slightly inclined guide plate has a better thermal performance than the full shield case. Furthermore, using a guide plate only with a small inclination angle has a superior thermal performance. Changing the position of the guide plate is investigated as well and the optimal location is obtained.

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## 1. Introduction

Thermal management of electronic components has been an active area of research and development focus due to the evolution of microelectronics over the last few decades. The increase in power density in applications such as microelectronic packages and aircraft electro-mechanical actuation (EMA) has pointed to the need for using effective cooling methods. These cooling methods help to keep the operating temperature of electronic components at an acceptable level. Much attention has been given recently on the challenges that are encountered in flight actuation

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systems such as high-power-density electric motors and power electronics. Failure to manage the heat generated in the EMA systems may lead to mission objective failure and detrimental loss of the aircraft [1,2]. McCarthy et al. [2] showed that the power electronics system has approximately 25% of the total heat load generated in a typical flight EMA system. At 200 W of heat load and without forced air cooling, the electronic substrate temperature can reach  $\sim 107^\circ\text{C}$  which exceeds the imposed temperature limit to avoid overheating of the power electronics. To assure the reliability of the electronics and to lower the operating and maintenance costs of the flight actuation systems, an air-cooling system is considered as the primary cooling method for EMA system components [3].

Several studies have been undertaken on heat sink design for improved cooling of electronic components that include steady-

**Nomenclature***Alphabet – Upper Case*

|           |                                             |
|-----------|---------------------------------------------|
| $A_t$     | total convection area of a heat sink, $m^2$ |
| $B$       | heat sink footprint width, mm               |
| $D_h$     | duct hydraulic diameter, m                  |
| $E$       | total energy per unit mass, J/kg            |
| $H$       | fin height, mm                              |
| $H_s$     | shield height, mm                           |
| $H_b$     | bypass height above the shield, mm          |
| $H_{BP}$  | total bypass height above the heat sink, mm |
| $H_p$     | guide plate height, mm                      |
| $L_{hs}$  | heat sink footprint length, mm              |
| $N$       | number of fins                              |
| $Nu_{Dh}$ | Nusselt number using $D_h$                  |
| $P_1$     | channel inlet pressure, N/m <sup>2</sup>    |
| $Q$       | heat dissipation rate, W                    |
| $R_{th}$  | total thermal resistance, K/W               |
| $Re_{Dh}$ | Reynolds number                             |
| $T_b$     | base plate temperature, K                   |
| $S$       | fin spacing, mm                             |

|       |                      |
|-------|----------------------|
| $V_d$ | duct velocity, m/s   |
| $X_p$ | guide plate position |

*Alphabet – Lowercase*

|       |                                                |
|-------|------------------------------------------------|
| $h$   | heat transfer coefficient, W/m <sup>2</sup> ·K |
| $f$   | friction coefficient                           |
| $t_f$ | fin thickness                                  |
| $t_b$ | baseplate thickness, mm                        |
| $q''$ | Heat flux, W/m <sup>2</sup>                    |
| $y^+$ | dimensionless wall distance.                   |

*Greek symbols*

|               |                                                              |
|---------------|--------------------------------------------------------------|
| $\varepsilon$ | turbulent dissipation rate, m <sup>2</sup> /sec <sup>5</sup> |
| $\mu$         | absolute viscosity, Pa·s                                     |
| $\rho$        | density, kg/m <sup>3</sup>                                   |
| $\theta_p$    | guide plate inclination angle, degree                        |
| $\Delta P$    | pressure drop, N/m <sup>2</sup>                              |

state and transient models, simple analytical and empirical models, and experimental approaches. Morega et al. [4] carried out a study to minimize the thermal resistance ( $R_{th}$ ) of a stack of parallel plate fin heat sink (PPFHS). The optimum heat transfer condition was determined for equal spacing between the plates. Furthermore, there is an optimal number of plates that minimizes the  $R_{th}$  for a specified free stream velocity when the overall dimensions of the PPFHS are specified. In comparison with other simple internal duct geometries, PPFHSs are thought to offer the best combination of high heat transfer and low pressure drop for high-performance heat sinks. This information was confirmed by the experimental results conducted by El-Sayed et al. [5] who examined the thermal/hydraulic behavior of a PPFHS exposed to parallel flow, impinging and reverse-impinging flow. They concluded that parallel flow has the highest rate of heat transfer and the lowest pressure drop.

The performance of a PPFHS with the presence of flow bypass has been reported in the literature by several researchers. For instance, Lau and Mahajan [6] determined experimentally the effects of fin density and tip clearance on the PPFHS thermal and hydraulic performance. Their data indicated that significant gains in thermal performance can be obtained with high fin density heat sinks for a moderate increase in pressure drop. Tip clearances should be kept to a minimum for high thermal performance. Wirtz et al. [7] experimentally derived a correlation for inter-fin velocity as a function of approach free stream flow rate and heat sink fin density. They showed that the overall heat transfer coefficient of these fins can be obtained using the analytical solutions presented in [8] for developing flow between parallel plates. Moreover, their results showed that 60% of the air flow may bypass the PPFHS depending on the fin density of the heat sink and the geometry of the flow duct. This high level of air bypass results in approximately 50% reduction in the heat sink performance. Jonsson and Moshfegh [9] developed empirical bypass correlations for a PPFHS to predict the pressure drop and Nusselt number with an accuracy of  $\pm 25\%$  and  $\pm 10\%$ , respectively. Leonard et al. [10] tested two PPFHSs with different channel/fin height ratios. A simple bypass model was used to estimate the average flow velocity inside the heat sinks with  $\pm 8\%$  agreement with experimental data. Elshafei [11] theoretically and experimentally investigated the effects of the flow velocity, fin density, tip-to-shroud clearance (flow bypass) on the thermal performance and pressure drop across aluminum

PPFHSs. Their results showed that the heat transfer rate decreases with an increase in the tip-to-shroud clearance. For instance, the thermal performance reduces by about 30% at a clearance ratio of 0.89 when compared to the fully shrouded case. Li and Chao [12] measured the thermal performance of PPFHSs with different fin density and flow velocity. They showed that the  $R_{th}$  decreases with the increase in the fin height and the optimal fin thickness was determined for different Reynolds numbers. Wu et al. [13] presented an analytical model to assess the thermal/hydraulic performance of a PPFHS with a wide range of Reynolds numbers. The model was able to predict the pressure drop within 8.4% to 13.9% compared to experimental results. Recently, Chingulpitak et al. [14] investigated the hydraulic/thermal performance of a PPFHS with cross-cutted fin. Their results indicated that fin cross-cutting can yield approximately a 16.2% decrease in the heat sink  $R_{th}$  for the same pumping power.

Computational fluid dynamics (CFD) approaches have been extensively applied to study heat transfer and flow behavior in heat sinks. Yuan [15] showed that the inter-fin velocity decreases with distance downstream which affects the heat transfer rate. Barrett and Obinelo [16] revealed that the augmentation in the heat sink heat transfer rate is due to surface area increase. Prstic et al. [17] numerically studied the bypass effects on the performance of a heat sink at different air flow rate, fin density and clearance area ratio. They concluded that the bypass flow increases with increase in fin density and clearance distance. Chen et al. [18] studied the performance of a PPFHS using an inverse method model and FLUENT software in conjunction with an experimental approach. Good agreement in the fin temperatures between the calculated values and the experimental data was obtained.

Recently, Ahmed et al. [19] reviewed different thermal and hydraulic optimization approaches for the PPFHS. Several studies were reported on the improvement of the performance of the PPFHS by promoting the turbulence of the flow through the inter-fin spacings [20,21]. For example, Yang et al. [20] studied experimentally the performance of PPFHS with different types of loose and dense vortex generators (VG). They concluded that vortex generators operating at a large fin pitch and a high frontal velocity are more effective than the plain PPFHS. The delta-winglet pair VG provides the optimal enhancement design as it can reduce by 12–15% the surface area at a frontal velocity of approximately 3–5 m/s. Ahmed et al. [21] optimized the thermal

design of a PPFHS by inserting ribs at different locations and with different numbers and orientation in the space between fins. They concluded that a ribbed PPFHS can increase the thermal performance and decrease the pressure drop compared to conventional PPFHS. Ahmed et al. [19] also reported some papers that studied the characteristics of the flow through PPFHS if vortex generators or flow tabulators are installed through the tip clearance. For instance, Li et al. [22] experimentally investigated the effect of vortex generators attack angle and separating distance between the PPFHS and vortex generator on the thermal/hydraulic performance of PPFHS. The superior performance in terms of lower  $R_{th}$  and lower pressure is recorded at an attack angle of  $30^\circ$  and zero separating distance. For the lowest  $Re$ , the result showed that the  $R_{th}$  of the PPFHS with the vortex generator is 27% lower than that of the regular PPFHS. Prstic and Bar-Cohen [23] compared the performance of a partially/fully shrouded and a shielded heat sink experimentally and numerically. The fully shrouded heat sink has the best thermal performance, but this comes with a high pressure drop penalty. Adding a heat shield to a partially shrouded heat sink decreases the pressure drop while maintaining a thermal resistance comparable to the fully shrouded configuration. Li et al. [24] tested the installation of the vertical shield on PPFHS and found that more coolant fluid is forced to pass through the fin spacing. This promotes heat transfer, but the pressure drop across the heat sink rises. Furthermore, Tsai et al. [25] investigated the effect of the different shield inclination angle ( $\theta$ ) on the thermal/hydraulic performance of a PPFHS. They found that the variation of this angle can cause a substantial and complicated variation to the flow field both upstream and downstream of the heat sink. Correlations to assess the thermal/hydraulic performance of the heat sink with a shield were developed by Shaalan et al. [26]. They concluded that the inclination angle yielding the minimum  $R_{th}$  varies with the heat sink height. Zhang et al. [27] studied the effects of shield position and height on the thermal/hydraulic performance of a PPFHS experimentally. Their experimental data revealed that superior thermal performance occurs when the shield is half the height of the fins and is positioned at the top front of the heat sink. They developed correlations for the average Nusselt number and friction factor and the predicted values fitted the experimental data with deviations less than  $\pm 15\%$ . Recently, Elsayed and Mesalhy [28] proposed a performance enhancement for a shield heat sink by making slots on the shield. The slots create jets that are able to eliminate the eddies and vortices generated by the solid shield. They reported the appropriate slot number and width to achieve the optimum thermal/hydraulic performance. For the highest  $Re$  number, a slotted shield offers a 80% lower pressure drop compared to a solid shield.

Most of the work available in the literature has focused on the effect of the geometrical variables of a PPFHS, such as fin height, fin thickness and fin density on the hydraulic/thermal performance. As the previous literature indicates, it is feasible to enhance the heat transfer of fin array by altering the bypass through the installation of a shield at different positions and inclination angles, and by forming slots in the solid shield. Shielding the entire bypass region produces an enhancement in heat transfer, but this causes a dramatic increase in the pressure drop at the same time. There are two main areas of investigation in this paper. The first is to assess the merit of using a partial shield with a guide plate to direct the bypass flow to impinge at the heat sink. Comprehensive performance evaluations of the heat sink with different partial shield heights and guide plate inclination angles are presented to find the optimum configuration that gives the maximum heat transfer rate with low pressure drop. General correlations for the hydraulic/thermal performance of the PPFHS with partial shield and guide plate are obtained and compared with experimental correlations available in the literature. To the best of the authors' knowledge,

none of the literature mentioned above includes a fan performance curve as the inlet boundary condition which is a far more common scenario than the fixed inlet approach velocity. Therefore, the second focus of the paper is on performing an optimization analysis for the different installations using an actual fan operational curve. This analysis will establish the direct connection between the optimized pressure drop and flow rate with the system operating curve of the selected fan.

## 2. Numerical approach

### 2.1. Problem description and mesh generation

The assembly for the wing bay with an EMA installed is shown in Fig. 1a. The EMA consists of four main components including a motor, drivetrain, gearbox and electronic box. The electronic box is simulated as a fixed heat source and the considered PPFHS with length  $L_{hs}$  (100 mm) and a baseplate thickness  $t_b$  of 5 mm is mounted at the bottom of the duct as shown in Fig. 1b. The figure

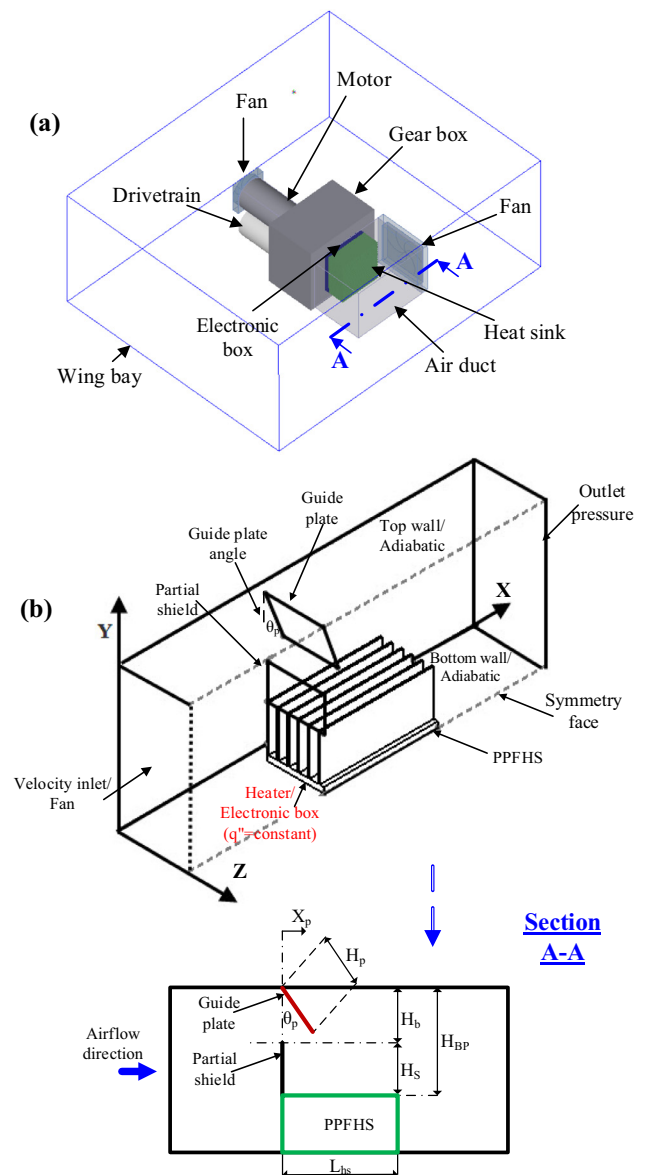


Fig. 1. (a) Wing bay with EMA assembly and (b) Boundary conditions for heat sink with a partial shield and guide blade.

also shows the boundary conditions for numerical simulation as well as the configurations of the PPFHS, partial shield and guide plate. Additional geometrical parameters are given in Table 1 as well. The total bypass height above the heat sink  $H_{BP}$  is set at 85 mm. The distance above the partial shield  $H_b$  can be determined as  $H_b = H_{BP} - H_s$ , where  $H_s$  is the height of the partial shield. The guide plate height  $H_p$  is equal to  $H_b$ . The assumptions made in the current numerical model include: steady-state operation, three-dimensional flow and heat transfer, temperature-independent thermophysical properties and negligible radiation heat transfer. These assumptions are valid in typical applications of EMA cooling. The upstream inlet and downstream exit are changed with respect to the heat sink length from  $0.5L_{hs}$  to  $3L_{hs}$ . It is noted that the variations in the maximum baseplate temperature ( $T_{b,max} - T_a$ ) for different upstream and downstream lengths are within 2% of the ( $T_{b,max} - T_a$ ) for  $0.5L_{hs}$  case. Therefore, for all the simulation cases in the current work, the air inlet and outlet lengths are set to  $0.5L_{hs}$ .

The temperature and flow fields were calculated numerically with the commercial CFD software ANSYS FLUENT 16 [29], under the following assumptions: steady state, incompressible, turbulent, constant thermophysical properties, and negligible gravitation and thermal radiation effects. The equations governing the fluid are the continuity equation, Reynolds-averaged Navier–Stokes equation and the energy equation. Based on the assumptions above these equations can be expressed as:

$$\frac{\partial}{\partial x_i}(\rho V_i) = 0 \quad (1)$$

$$\frac{\partial}{\partial x_i}(\rho V_i V_j) = -\frac{\partial P}{\partial x_i} + \frac{\partial \tau_{ij}}{\partial x_j} \quad (2)$$

$$\frac{\partial}{\partial x_i}[V_i(\rho E + P)] = \frac{\partial}{\partial x_i}\left(k \frac{\partial T}{\partial x_i}\right) \quad (3)$$

where  $i$ : is a tensor indicating 1, 2, and 3 and  $\tau_{ij}$  is the viscous stress tensor.

The RNG (Reynolds Normalized Group)  $\kappa$ - $\epsilon$  turbulent model was used to solve the complex turbulent thermal flow field with enhanced wall function approach and viscous heating in the near-wall regions to fit the wall boundary conditions [29]. Eqs. (4) and (5) are for transport equations for the turbulence kinetic energy ( $k$ ) and rate of dissipation  $\epsilon$ , respectively. The numerical solution is assumed to have converged when the scaled residual of the energy equation reached  $10^{-7}$ , and the scaled residuals of other equations reached  $10^{-4}$ .

$$\frac{\partial(\rho k u_i)}{\partial x_i} = \frac{\partial}{\partial x_j}\left(\alpha_k \mu_{eff} \frac{\partial k}{\partial x_j}\right) + G_k - \rho \epsilon \quad (4)$$

**Table 1**  
Values of geometrical and thermal parameters considered.

| Value              | Parameter                                            |            |
|--------------------|------------------------------------------------------|------------|
| 25                 | Air inlet temperature ( $T_a$ ), °C                  |            |
| 30                 | Heat Input ( $Q$ ), W                                |            |
| 1, 3, 5 and 7      | Duct air inlet velocity ( $V_a$ ), m/s               |            |
| 125                | Duct hydraulic diameter ( $D_h$ ), mm                |            |
| 35                 | Fin height ( $H$ ), mm                               | PPFHS      |
| 1.5                | Fin thickness ( $t_f$ ), mm                          | Dimensions |
| 20                 | Fin number ( $N$ )                                   |            |
| 5                  | Baseplate thickness ( $t_b$ ), mm                    |            |
| 100                | Width ( $W$ ), mm                                    |            |
| 100                | Length ( $L_{hs}$ ), mm                              |            |
| 3.68               | Fin spacing ( $S$ ), mm                              |            |
| 85                 | Total bypass height ( $H_{BP}$ ), mm                 |            |
| 20, 30, 40, 50, 60 | Partial shield height ( $H_s$ ), mm                  |            |
| 15, 25, 35, 45, 55 | Guide plate inclination angle ( $\theta_p$ ), degree |            |

$$\frac{\partial(\rho \epsilon u_i)}{\partial x_i} = \frac{\partial}{\partial x_j}\left(\alpha_\epsilon \mu_{eff} \frac{\partial \epsilon}{\partial x_j}\right) + C_{1\epsilon} \frac{\epsilon}{k} (G_k) - C_{2\epsilon} \rho \frac{\epsilon^2}{k} - R_\epsilon \quad (5)$$

In these equations,  $G_k$  is the generation of  $k$  due to the mean velocity gradients. The quantities  $\alpha_\epsilon$  and  $\alpha_k$  represent the inverse effective Prandtl number for  $\epsilon$  and  $k$ , respectively. The production term is represented by  $R_\epsilon$ . The model constants  $C_{1\epsilon}$  and  $C_{2\epsilon}$  are 1.42 and 1.68, respectively [29].

The Hex/Sub and hex/wedge meshing schemes are used in the current study for the computational domain with mesh refinement near the walls and along sharp or rough edges. To examine the dependence of the present numerical simulation results on the grid density, simulations were carried out with various mesh grid numbers in the X, Y and Z directions. As recommended by FLUENT [29], the grid is refined near the walls that are in contact with air in order to keep the  $y^+$  values for the points approaching the walls between 1 and 5. The grid sensitivity analysis was performed to assure the grid independence of the results. When the grid number was increased above 740,000, only a 0.02% variation in thermal resistance was obtained.

## 2.2. Thermal and hydraulic performance parameters

In this section, the methods of calculating the parameters needed to evaluate the thermal and hydraulic performance of the PPFHS are described. The thermal performance of the PPFHS is evaluated by calculating the thermal resistance which is defined as:

$$R_{th} = \frac{T_{b,max} - T_a}{Q} \quad (6)$$

where  $T_{b,max}$  is the maximum baseplate temperature,  $T_a$  is the inlet temperature of the cooling air, and  $Q$  is the heat dissipation rate by the PPFHS.

The heat transfer coefficient for the PPFHS is obtained using the Newton's law of cooling:

$$h = \frac{Q}{A_t(T_{b,max} - T_a)} = \frac{1}{R_{th} A_t} \quad (7)$$

The total heat transfer area ( $A_t$ ) is calculated by  $A_t = NL_{hs}(2H + t_f) + (N - 1)SL_{hs}$ .

The Reynolds number is given by:

$$Re_{Dh} = \frac{\rho V_a D_h}{\mu} \quad (8)$$

where  $D_h$  is the hydraulic diameter of the flow cross-section, and  $\mu$  is the dynamic viscosity of air.

The Nusselt number is calculated as:

$$Nu_{Dh} = \frac{h D_h}{k_f} \quad (9)$$

where  $k_f$  is the thermal conductivity of the air.

The friction factor is defined as:

$$f = \frac{2\Delta P}{\rho(L/D_h)V_d^2} \quad (10)$$

where  $\Delta P$  is the pressure drop across the PPFHS,  $L$  is the distance between the upstream and downstream pressure ports. The air thermophysical properties used in the parameters calculation are based on the average temperature of the air flow between the inlet and outlet test section.

## 2.3. Model validation

To validate the computational model using the FLUENT software, a comparison between the numerically calculated perfor-

mance parameters and actual experimental measurement data available in the literature is made and shown in Fig. 2. Several experiments on a PPFHS were done by Shaalan et al. [26]. The PPFHS is similar to the heat sink in the present study, with  $N = 12$  fins,  $H_s = 40$  mm and shield angle equals to  $90^\circ$  at various Reynolds numbers. Using the reported relative uncertainty in the experimentally measured parameters, the calculated uncertainties in  $Nu_{Dh}$  and  $fRe_{Dh}$  are 2.01% and 3.97%, respectively. Fig. 2 shows good agreement between the current numerical model calculations and the experimental measurements. The calculated root mean square discrepancies between the experimental data and numerical values of  $Nu_{Dh}$  and  $fRe_{Dh}$  are 2.2 and 5.3%, respectively. With the current model validated, computations were conducted for a heat sink with a partial shield and guide plate with different inclination angles.

### 3. Results and discussion

Two different scenarios of forced convections are considered. For the first scenario, the volume flow rate is fixed at the inlet of the duct. Two sets of results are studied. In the first set, the performance of the heat sink with a partial shield is tested for different values of open area ratio. In the second set, the effect of inserting the guide plate with at different inclination angles is shown. For the second scenario, air flow is driven by a fan with the fan curve specified at the inlet. The thermal performance of the PPFHS under different partial shield and guide plate configurations is studied and compared with the full shield configuration.

#### 3.1. Partial shield with different open area ratios

The open area ratio is defined as the ratio between the allowed open area above the partial shield and the total bypass area:

$$\beta = \frac{H_b}{H_b + H_s} = \frac{H_b}{H_{BP}} \quad (11)$$

For the case of the full shield, the values of  $H_b$  and  $\beta$  are both zero. With full open bypass the values of  $H_s$  and  $\beta$  are 0 and 1, respectively. To get a better understanding of the effect of using a partial shield with different  $\beta$  on the hydraulic/thermal performance of PPFHS, the Nusselt numbers and friction factors are

determined for different Reynolds numbers and depicted in Fig. 3. It is clear that the Nusselt numbers for the full shield case are remarkably larger than those for the fully-open configuration (without a shield) for all  $Re_{Dh}$  numbers. The variation of the  $fRe_{Dh}$  with  $Re_{Dh}$  is relatively small for the cases without a shield and short partial shields. The  $fRe_{Dh}$  increases with decreasing open area  $\beta$  for cases with the partial shield installation till it reaches a maximum value in the full-shield case. This is attributed to the increase in the frictional losses as the Reynolds number increases and the open area ratio decreases. This conclusion is similar to the results reported in [24–27].

#### 3.2. Partial shield with guide plate

##### 3.2.1. Flow streamlines

The velocity contours, pressure contours and streamlines at  $Re = 55,000$  for the full-shield case and the partial shield with guide plate case ( $H_s/H_{BP} = 0.23$  and  $\theta_p = 15^\circ$ ) at the plane of symmetry that passes between two adjacent fins are portrayed in Fig. 4. For the full-shield case, the streamlines show that the blockage by the solid shield forces the cooling air to flow through the PPFHS and increases the air velocity between the fin channels. Nearly all the mass flow rate passes through the fin channels which enhances heat transfer. On the other hand, the blockage effect of the shield increases the upstream pressure which in turn increases the pumping power required for the cooling process. Two vortices are formed in the full shield case as shown in Fig. 4. The first vortex, which is the more significant one, is formed behind the shield while the smaller one is formed after the flow passes the heat sink. Reducing the blocked area can be achieved by using a partial shield only which in turn allows more fluid to bypass the heat sink and alleviate the high pressure formed upstream. It is clear that, adding a guide plate to the partial shield can direct the bypass flow to impinge onto the heat sink top surface and add an extra cooling effect as shown in the velocity contours in Fig. 4. From the pressure contours, the considerable difference in the upstream pressure between the two cases can be clearly noticed. Also, from the streamlines shape of the partial-shield with guide plate case, the flow is rather complex. Two vortices are formed behind the partial shield and one additional vortex is created behind the guide plate. The vortex behind the guide plate decreases as the guide plate inclination angle increases which lead to a lower pressure drop.

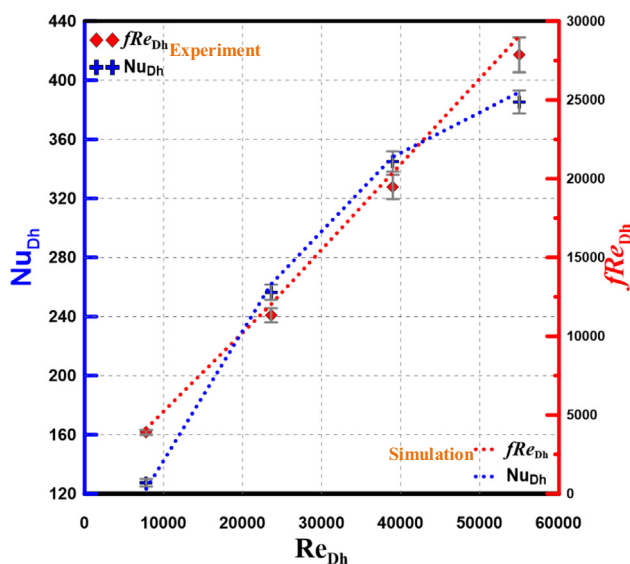


Fig. 2. Comparison of simulation with experimental data at different Reynolds numbers [26].

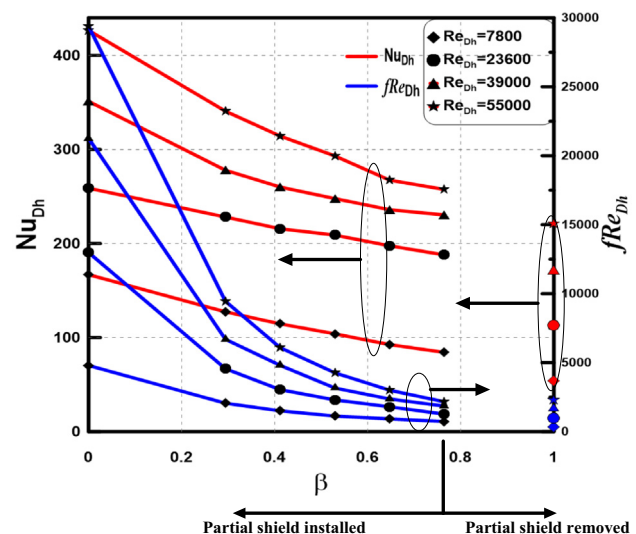


Fig. 3. Variation of Nusselt numbers and friction factors with different  $Re_{Dh}$ , and  $\beta$  for partial shield cases.

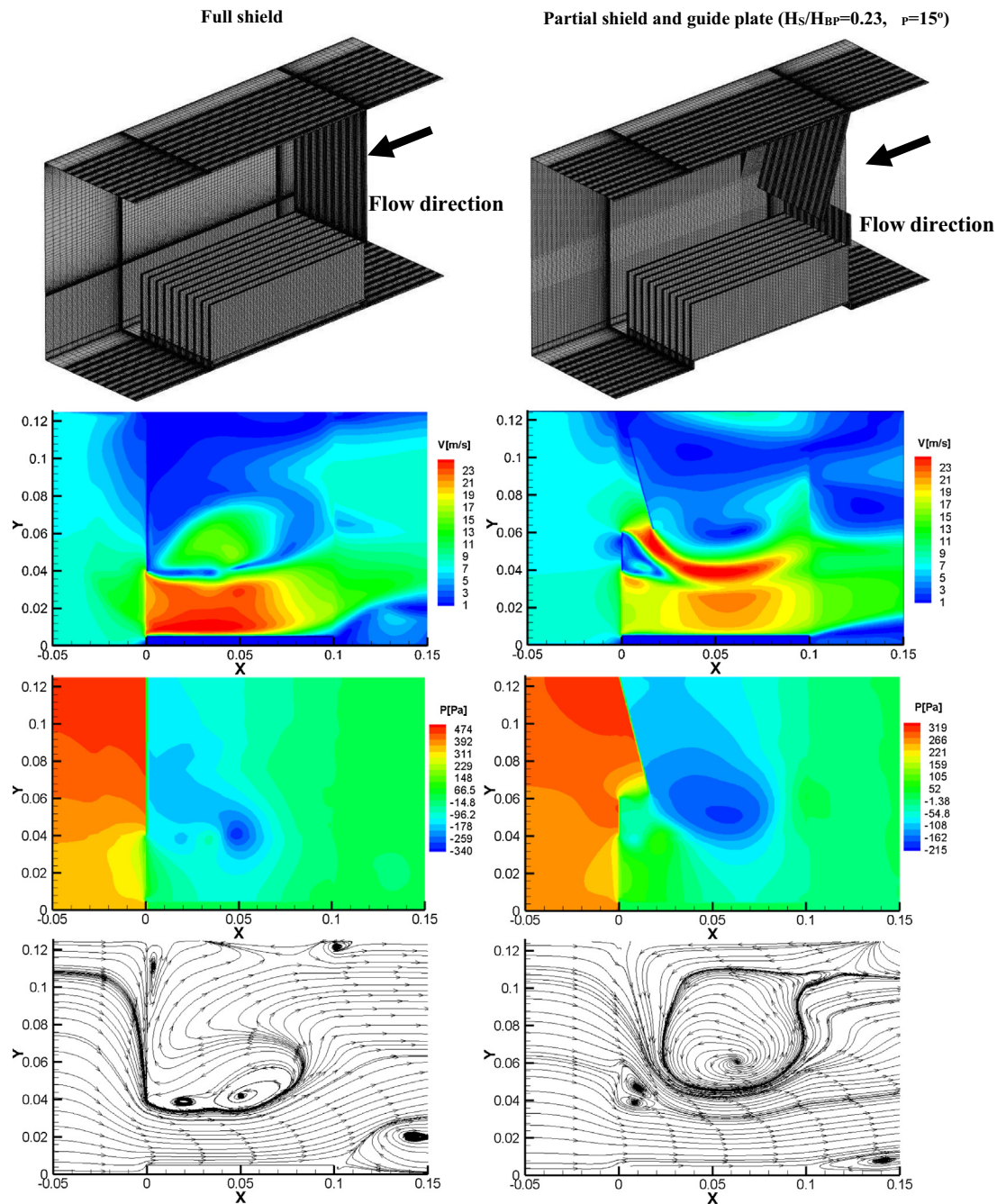


Fig. 4. Computational domains, velocity contours, pressure contours and streamlines at symmetry plane for  $Re = 55,000$ . Axis dimensions are in m.

On the other hand the vortices behind the partial shield increase with increase in the partial shield height till they reach the size formed in the full shield case.

### 3.2.2. Effect of shield height and guide plate inclination angle

The effect of the guide plate and partial shield combination on the thermal and flow performance of the heat sink operating at two different values of Reynolds number is shown in Fig. 5. It is worth to note that the shield height plus the distance above the shield is constant, which is the total bypass distance ( $H_{BP} = 85$  mm). So as the shield height decreases the guide plate gets closer to the heat sink. For most of the cases, the pressure drop reaches its minimum value when  $H_s/H_{BP}$  ranges approximately between 0.35 and 0.5. For higher values of this ratio ( $H_s/H_{BP} > 0.5$ ) the pressure drop is high due to the blockage effect of the partial shield. On

the other hand, for lower values ( $H_s/H_{BP} < 0.35$ ) the pressure drop increases due to the flow impingement with the heat sink and the steep turn of the flow around the guide plate and the increased blockage of the guide plate. For the lower Reynolds numbers, similar trends are observed for Nusselt number for  $\theta_p$  equal to  $25^\circ$  or higher while for the higher Reynolds numbers, Nusselt number increases or stay nearly constant as the ratio  $H_s/H_{BP}$  increases. Also, the figure shows that both the friction factor and Nusselt number decrease significantly with increasing  $\theta_p$  for all  $H_s/H_{BP}$  cases and for both Reynolds numbers. The case of  $\theta_p = 15^\circ$  has the highest thermal performance and the highest pressure drop. So, choosing the favorable operating conditions for the partial shield with guide plate should be decided by optimizing the shield height and the guide plate inclination angle that produces a higher heat transfer effect with a lower pressure drop penalty.

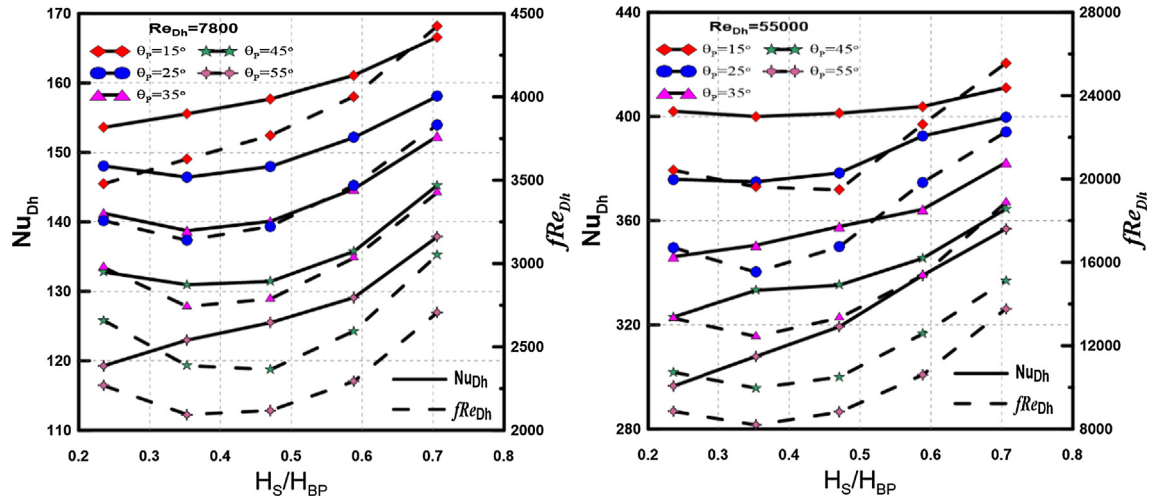


Fig. 5. Nusselt numbers and friction factors versus partial shield height at different guide plate angles.

### 3.3. Empirical correlation

According to the above discussion, the average Nusselt number and the friction factor are dependent on the Reynolds number,  $H_S$  and the inclination angle  $\theta_p$  when a guide plate is used. Considering the previous parameters, correlations for  $Nu_{Dh}$  and  $fRe_{Dh}$  are derived and presented in Table 2 for both partial shield with and without guide plate. The deviation in Nusselt number correlations is about  $\pm 13\%$  for the partial shield without guide plate configuration and  $\pm 8\%$  for the partial shield with guide plate configuration. Also, the maximum deviation in the friction factor correlations is about  $\pm 17\%$  for both cases. Table 2 shows all the constants and exponents (parameters) estimated for the  $Nu_{Dh}$  and  $fRe_{Dh}$  correlations for both cases of the partial shield and partial shield with the guide plate. The values of  $R^2$  confirm the statistical goodness of the fit. The obtained correlations are applicable in the ranges mentioned. Also, the performances of the correlations for different cases are shown in Fig. 6.

The currently proposed heat transfer correlation for the specified PPFHS ( $N = 20$  fins and  $H = 40$  mm), Eq. (12), is compared with an experimental-based correlation, Eq. (16), for flow over a heat sink with a right angle shield given in [26].

$$Nu_{Dh} = 1.17(Re_{Dh})^{0.6} + 0.588 \left( Re_{Dh} \frac{H}{C_H} \right)^{-0.018} \left( \frac{S}{C_H} \right)^{-0.067} \times \left( \frac{\theta}{\theta_{90}} \right)^{-1.25} \begin{cases} 7800 \leq Re_{Dh} \leq 5.5 \times 10^4 \\ 0.16 \leq \frac{H}{C_H} \leq 0.64 \\ 0.16 \leq \frac{H}{C_H} \leq 0.64 \\ 0.52 \leq \theta(rad) \leq 1.57 \end{cases} \quad (16)$$

Table 2  
Correlations for partial shield only and partial shield with guide plate combination.

| Correlations Equations                                                                                                                    | Range and case type                                                             | $R^2$ | Error |      |
|-------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|-------|-------|------|
| $Nu_{Dh} = 1.68(Re_{Dh})^{0.5} \left( \frac{H_{BP}}{H_s} \right)^{-0.336}$                                                                | Partial shield<br>$7800 \leq Re_{Dh} \leq 5.5 \times 10^4$                      | 0.96  | 13%   | (12) |
| $fRe_{Dh} = 0.96(Re_{Dh})^{0.95} - 0.463(Re_{Dh})^{0.993} \left( \frac{H_{BP}}{H_s} - 1 \right)^{0.137}$                                  | $0 \leq \frac{H_{BP}}{H_s} \leq 4.25$                                           | 0.99  | 17%   | (13) |
| $Nu_{Dh} = 2.31(Re_{Dh})^{0.48} \left( 1 + \frac{\theta_p}{\theta_{90}} \right)^{-0.602} \left( \frac{H_{BP}}{H_s} - 1 \right)^{-0.041}$  | Partial shield and guide plate<br>$7800 \leq Re_{Dh} \leq 5.5 \times 10^4$      | 0.99  | 8%    | (14) |
| $fRe_{Dh} = 1.83(Re_{Dh})^{0.89} \left( 1 + \frac{\theta_p}{\theta_{90}} \right)^{-2.273} \left( \frac{H_{BP}}{H_s} - 1 \right)^{-0.162}$ | $0 \leq \frac{H_{BP}}{H_s} \leq 4.25$<br>$15^\circ \leq \theta_p \leq 55^\circ$ | 0.98  | 16%   | (15) |

where  $H$ ,  $C_H$  and  $S$  are the fin height, channel height and fin spacing, respectively. To give the same heat sink dimensions used in the present study, the values of 0.28 and 0.0295 are used for  $H/C_H$  and  $S/C_H$ , respectively. The difference between the obtained results from the proposed correlation Eq. (12) and the results from the correlation Eq. (16) reported in [26] is within 4%, which shows a good agreement.

### 3.4. Performance using an operating fan

In the previous sections, optimization analysis and comparison between different partial shield configurations and partial shield/guide plate combinations are based on a fixed free stream inlet velocity. The inlet velocity approach assigns different inlet velocity values independently without considering the head loss produced. While such similar efforts were reported in numerous papers in the literature as discussed earlier, real applications are less likely to have a constant volume flow rate fan. A more realistic situation is to place a fan that adapts an inverse relationship between head loss and volumetric flow rate [30]. As the pressure drop across the bypass flow, the volumetric flow rate and in turn the inlet stream velocity decreases. A comparison of thermal performance based on a fixed inlet velocity is therefore not appropriate. Perhaps a better comparison can be made by using an actual fan curve as the inlet boundary condition [31]. The analysis approach in the current section establishes the connection between the thermal performance of a PPFHS with different shield and guide plate installations as in the previous section using the system operating curve for a selected fan. A commercially available fan (Sunon PF92381) is chosen for the current simulation based on the size requirements for the wing bay, rotating speed range and the antic-

ipated volume flow rate and pressure head requirements. The operating fan curve supplied from the manufacturer for the selected fan at its highest speed of 12,000 rpm is used with the fan operating at 12 V. The optimization approach used in the pre-

vious sections is repeated with the fan curve mentioned above set as the inlet boundary condition. In the numerical solution model, the inlet fan boundary condition is defined as a fan pressure jump at the ambient pressure and specific rotational speed from the

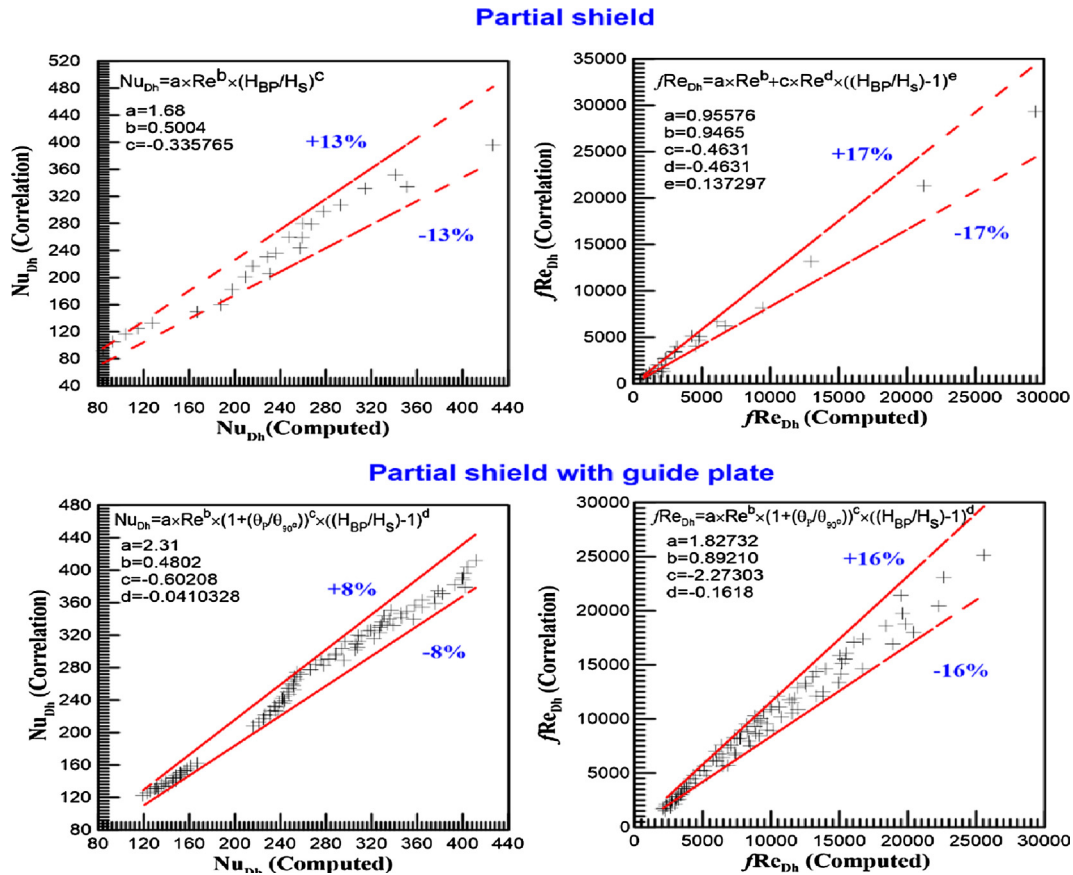


Fig. 6. Correlations performances obtained Nusselt number and friction factor versus computed values.

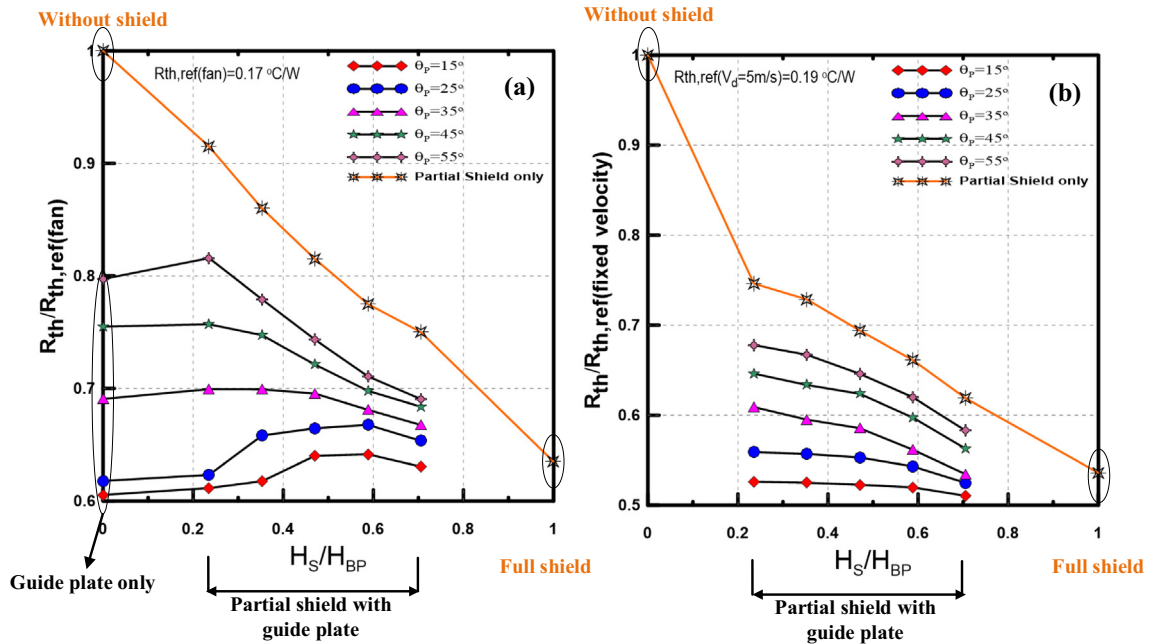


Fig. 7. Thermal performance enhancement due to partial heat shield installation and different guide plate inclination angles for (a) fan curve and (b) fixed velocity 5 m/s.

manufacturer webpage. The fan pressure jump is specified as a third order polynomial function of the fan volume flow rate.

In this part of the current work, the thermal resistance is used to define the system thermal performance. Also, the thermal and hydraulic operating conditions of PPFHS without any installations is selected as the reference case in the present calculations. The calculated reference thermal resistances for the inlet fan and the fixed inlet velocity boundary conditions are 0.17 and 0.19 °C/W, respectively. The different installation combinations are compared with the reference cases to assess the overall thermal performance enhancement of the PPFHS. The variation of  $R_{th}/R_{th,ref}$  with  $H_s/H_{BP}$  for partial shield only and partial shield/guide plate combination is shown in Fig. 7a and b for fan inlet boundary condition and fixed velocity inlet, respectively. It is worth to mention that the thermal resistance ratio  $R_{th}/R_{th,ref}$  is the reciprocal of  $Nu_{Dh}/Nu_{Dh,ref}$ . Therefore, a value of  $R_{th}/R_{th,ref}$  lower than 1 implies that the overall ther-

mal performance is enhanced compared to the reference case. For the cases of guide plate only or PPFHS without any partial shields, the value of  $H_s/H_{BP}$  is considered to be equal to zero. For the installed shield only cases, the highest thermal performance for the PPFHS is attained when the lowest amount of air flow bypasses the heat sink (full shield case  $\beta = 0$  or  $H_s/H_{BP} = 1$ ). For the partial shield with guide plate combination, some important characteristics for different inclination angles can be concluded. For instance, If  $\theta_p \geq 35^\circ$ , decreasing the partial shield height raises the value of  $R_{th}/R_{th,ref}$  which means lower thermal performance. On the other hand, a different situation occurs for  $\theta_p < 35^\circ$  where the value of  $R_{th}/R_{th,ref}$  increases as  $H_s$  increases but then decreases with further increase in  $H_s$ . Finally the  $R_{th}/R_{th,ref}$  attains a minimum value for the shortest partial shield ( $H_s = 20$  mm) and guide plate angle  $15^\circ$ . In this case, the thermal performance is better than the full shield case as shown in Fig. 7a which is attributed to the air flow directed by the guide plate onto the surface of the PPFHS.

In case of guide plate only cases, the guide plate is anchored at the duct ceiling and can be positioned at different inclination angles with a length equals the bypass height above the heat sink. Fig. 7a shows that using a guide plate with a height of  $H_p = 85$  mm only and the inclination angle of  $\theta_p = 15^\circ$  offers the best thermal performance compared to all tested cases. The air gap between the guide plate and the heat sink directs the bypass air to impinge onto the PPFHS top surface. The selected fan supplies air for all tested cases with an average velocity in the range between 4.2 and 5.4 m/s. To compare with a fixed velocity boundary condition, an inlet air velocity of 5 m/s ( $Re = 39,000$ ) is chosen. For the case of a fixed flow rate, the thermal resistance decreases monotonically with increasing the partial shield height because the inlet velocity is kept constant regardless the change in pressure drop. However, for the case of fan inlet boundary condition, the flow rate decreases as the pressure drop increases. With fixed velocity inlet boundary condition, the minimum thermal resistance occurs at the largest partial shield height of  $H_s/H_{BP} = 0.7$  as shown in Fig. 7b. Both inlet boundary conditions show that the  $\theta_p = 15^\circ$  case has the highest thermal performance at any given  $H_s/H_{BP}$  value. But the present analysis with the boundary condition associated with the selected fan to the PPFHS shows other optimal geometrical configurations

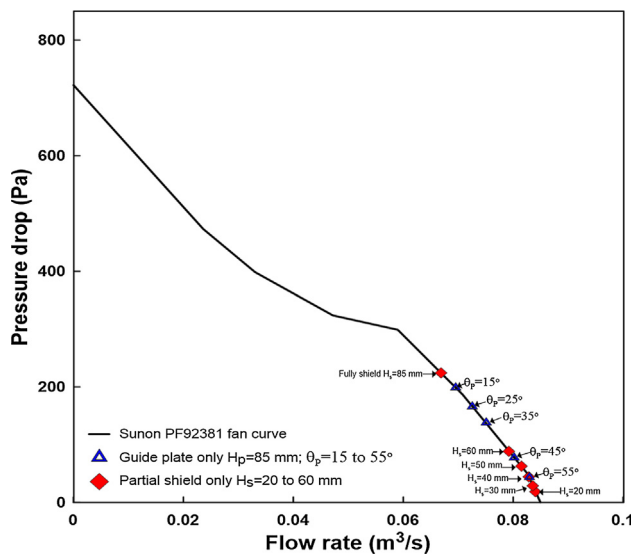


Fig. 8. System operating points for various guide plate and partial shield combination along with the fan curve.

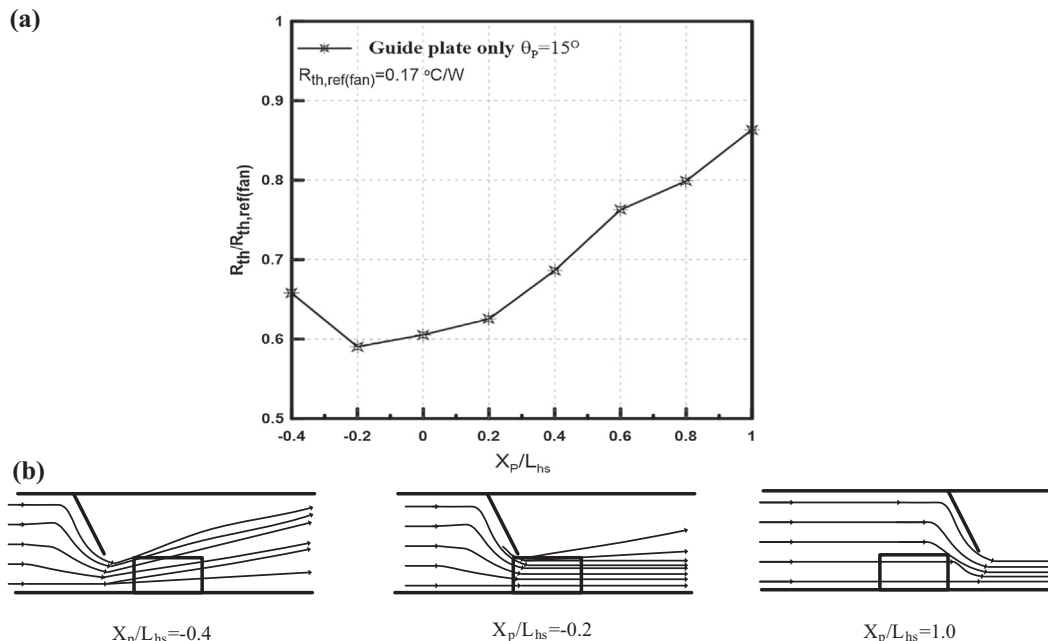


Fig. 9. (a) Effect of guide plate with  $\theta_p = 15^\circ$  on the thermal performance of the PPFHS. (b) Flow direction for selected guide plate location.

as shown in Fig. 7a. It should be noted that the results with fan inlet boundary condition are based on one characteristic curve of the selected fan. The results may change at different fan rotational speeds and different ambient pressures (different altitudes).

The calculated flow rate and the pressure drop (system operating point) for the installed heat sink with different partial shield or guide plate combinations together with the Sunon operating fan curve supplied by the manufacturer are shown in Fig. 8. For the partial shield cases, the shorter shield height, the higher is flow rate and the lower is pressure drop. For the guide plate only cases, the larger the inclination angle, the lower is the pressure drop and the higher is the flow rate.

Different positions for the guide plate are proposed and tested to study the effect of the guide plate location on the system thermal performance. Fig. 9a depicts the effect of the guide plate ( $\theta_p = 15^\circ$ ) dimensionless position  $X_p/L_{hs}$  on the thermal resistance ratio  $R_{th}/R_{th,ref}$ . The increase in the dimensionless value  $X_p/L_{hs}$  implies that the guide plate moves in the direction of the flow. As shown in the figure, the  $R_{th}/R_{th,ref}$  ratio increases as the  $X_p/L_{hs}$  increases which is attributed to the increase in the flow that bypasses the PPFHS partially which lowers the overall heat transfer. Further increase in the  $X_p/L$  exposes the PPFHS to more air bypassing the heat sink and causes more deterioration in the thermal enhancement ratio. On the other side shifting the guide plate against the flow direction lowers the  $R_{th}/R_{th,ref}$ . However, the ratio increases if  $X_p/L_{hs} < -0.2$ . The first decrease in  $R_{th}/R_{th,ref}$  is due to the maximum use of the air bypass obstruction with the guide plate. In this case, the air with lower temperature is obstructed and constrained to flow faster through the inter-fin channels. The reason for the increase in  $R_{th}/R_{th,ref}$  when  $X_p/L_{hs} < -0.2$  is because of the increasing gap for the air to bypass the heat sink. The approach flow will take the least resistance path below the guide plate trailing edge as shown in Fig. 9b. So, it is recommended to use a low plate inclination angle and position the guide plate so that the end of the guide plate points at the starting point of the heat sink.

#### 4. Conclusions

The thermal and hydraulic performance of PPFHS is investigated for proper thermal control of the electro-mechanical actuator (EMA) electronics. The inlet conditions adopted are a fixed mass flow rate and a selected fan characteristic curve. For the fixed flow rate inlet condition, the effects of the partial shield height, partial shield and guide plate combinations are studied to obtain the optimal configuration in terms of thermal and hydraulic performance. The inlet velocity cases are repeated for the fan inlet condition and the guide plate only installation is proposed as a possible approach. Based on the detailed results of the current study, the following conclusions can be drawn:

- Using only a partial shield decreases the large pressure drop caused by the full shield blockage, but this is accompanied by an increase in the thermal resistance. The Nusselt number and the friction factor increase with increase in Reynolds number and decrease in the dimensionless open area ratio.
- Associating a guide plate with a partial shield directs the bypass air to impinge onto the heat sink and consequently adds an extra cooling effect with a moderate increase in the pressure drop.
- The pressure drop reaches its minimum value for  $H_s/H_{BP}$  between 0.35 and 0.5. The same trend in Nusselt number occurs for low Reynolds numbers for guide plate inclination angles  $\theta_p = 25, 35$  and  $45^\circ$ , while for high Reynolds numbers, the Nusselt number decreases or stays nearly constant as the ratio  $H_s/H_{BP}$  decreases.
- The friction factors and Nusselt numbers decrease significantly with increasing  $\theta_p$  for all cases and all Reynolds numbers. For guide plate at  $\theta_p = 15^\circ$ , the highest thermal performance and the highest pressure drop are achieved.
- Correlations of the average Nusselt number and friction factor are fitted based on the governing dimensionless parameters ( $Re_{Dh}$ ,  $H_{BP}/H_s$ ,  $\theta_p/\theta_{90^\circ}$ ). The predicted values have a maximum deviation for the Nusselt number less than  $\pm 17\%$  and the maximum deviation for the friction factor less than  $13\%$ .
- When a Sunon fan curve is employed as the inlet condition, the results show a shorter shield with a small guide plate inclination angle has a better thermal performance compared to the full shield case.
- Using a guide plate only with a small inclination angle has the best thermal performance among all the tested cases.
- The effect of the guide plate location is investigated and the optimum location in terms of thermal performance is obtained where the dimensionless guide plate position is  $X_p/L_{hs} = -0.2$ .

#### Conflict of interest

Authors declare that there is no conflict of interest.

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